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Field Evaluation of Non-Radioactive Multiphase Flow Meters: Performance Comparison, Accuracy and Reliability Assessment

R. Almurayshid, H. Alshammari, E. Sadikhzada, and M. Alquraishi, Saudi Aramco; N. Mehendi, Weatherford; H. Gamal, Weatherford Saudi Arabia

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Abstract

Non-nuclear MPFMs offer an alternative to traditional nuclear-based flow metering systems, addressing environmental and regulatory concerns while potentially maintaining measurement accuracy and reliability. The objective of this study is to field test a non-nuclear multiphase flow meter (MPFM) and assess its performance under challenging operating conditions.

A non-nuclear MPFM consisting of Sonar, Near Infra-red, and Venturi was deployed in one of the stimulated wells during flowback operations to evaluate its capability to measure flow rates of oil, water, and gas phases accurately. The MPFM data will be compared to a downstream conventional separator to analyze the measurement accuracy, repeatability, and operational reliability under the same flowing conditions. Then the overall performance of non-nuclear MPFM will be compared to the nuclear MPFM. The PVT model was built based on the equation of state to convert the flow rates from operating to standard conditions.

The non-nuclear MPFM is anticipated to demonstrate reliable performance in measuring multiphase flow rates of oil, water, and gas during flowback operations. Preliminary results are expected to show measurement accuracy within an acceptable margin of error compared to the conventional separator data, with good repeatability under varying flow conditions. The Sonar, Near Infra-red, and Venturi technologies are likely to provide stable readings despite the complex fluid dynamics in the tight carbonate plays. Additionally, the comparison with nuclear MPFM may highlight advantages such as environmental safety, reduced regulatory burden, and comparable accuracy, making the non-nuclear MPFM a viable alternative for field applications.

This Paper discusses a novel non-nuclear flow measurement technology application. The authors will share recommendations and lessons learned from the comparison tests of MPFMs and separator during different flowing conditions.

Introduction

The global energy landscape is undergoing a profound transformation, driven by an escalating demand for hydrocarbons amidst increasingly complex and remote extraction environments. At the heart of efficient and sustainable oil and gas production lies the ability to accurately and continuously monitor oil, water,

and gas flow rates flowing from the wellhead. This is the domain of multiphase flow metering (MPFM), a critical technology that has evolved from rudimentary periodic well testing to sophisticated, real-time diagnostic systems. Accurate measurement of multiphase flow (oil, water, and gas) is critical for optimizing hydrocarbon production, reservoir management, and well performance evaluation.

Traditional methods, often relying on complex and capital-intensive test separators, presented significant limitations: delayed insights, increased operational footprint, substantial environmental impact, and a reactive approach to reservoir management. Over the past few decades, multiphase flow meters (MPFMs) have emerged as a viable alternative, offering continuous, real-time measurements at the wellhead. These devices reduce the need for frequent well testing, minimize flaring, and provide immediate insights into well performance (Zargar et al., 2021).

However, the journey of MPFM technology has not been without its challenges. Historically, a reliance on nuclear sources for phase fraction determination, while effective, introduced engineering and application challenges (Zargar et al., 2021). Non-nuclear MPFMs are gaining traction due to tightening safety regulations and advancements in hybrid sensor systems (Ramakrishnan et al., 2023).

Key Challenges in Multiphase Flow Metering for Oil & Gas

The accurate quantification of individual phase flow rates within a multiphase stream (oil, water, and gas) at the wellhead represents a cornerstone of efficient hydrocarbon asset management. This seemingly straightforward objective, however, is fraught with formidable technical and operational challenges, rendering multiphase flow measurement (MPFM) one of the most intricate challenges in upstream oil and gas operations. Unlike the relative simplicity of single-phase metering, the simultaneous presence and dynamic interaction of chemically distinct and physically heterogeneous fluids introduce a cascade of complexities that directly impact the precision, reliability, and economic viability of measurement solutions (Bruvik and Hjertaker, 2016; Mohindru, 2023).

Foremost among these difficulties is the pervasive issue of flow regime variability as shown in Figure 1. As hydrocarbons traverse the production tubing and flowlines, the distinct phases arrange themselves into a myriad of spatial configurations, each dictated by prevailing pressure, temperature, flow rates, pipe geometry, and fluid properties. These arrangements, known as flow regimes, span a spectrum from finely dispersed bubbly or mist flows to highly segregated stratified patterns, and critically, to inherently transient and chaotic slug flows. In bubbly flow, gas exists as discrete bubbles within a continuous liquid phase, presenting a relatively stable density profile (Zhao et al., 2023). Conversely, slug flow, characterized by alternating large plugs of liquid (often laden with entrained gas) and elongated gas pockets (carrying entrained liquid droplets), introduces abrupt and dramatic fluctuations in mixture density, velocity, and pressure. Annular flow sees liquid adhering as a film to the pipe wall while gas occupies the core, demanding different measurement approaches compared to stratified flow, where gravity dictates distinct layers of liquid and gas. Transitional flow, a tumultuous transition often observed between slug and annular regimes, defies stable characterization, posing an exceptional challenge for any measurement device. The fundamental problem is that most individual measurement principles, whether based on density, electrical properties, or acoustics, are acutely sensitive to these spatial distributions. A sensor calibrated for a homogenous bubbly flow will inevitably yield erroneous data when confronted with a violent slug, necessitating sophisticated algorithms and, ideally, multi-modal sensor fusion to accommodate this inherent variability (Kumar, 2023).

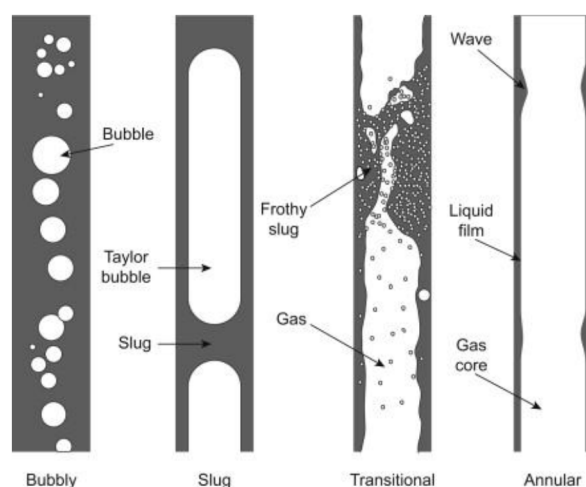


Figure 1—Flow Regime ((Bahadori, 2014))

Compounding the challenge of flow regimes are the complex and often unpredictable variations in fluid properties themselves. Hydrocarbon streams are rarely static in their composition. The specific gravity of crude oil can range dramatically, influencing its density, which further changes with dissolved gas content. Produced water salinity, a critical parameter for electrical property-based measurement techniques, can fluctuate over time, particularly as reservoir characteristics evolve or formation waters breakthrough (Almurayshid et al., 2024). Viscosity, a measure of fluid resistance to flow, plays a pivotal role in dictating flow regime transitions and can vary significantly with temperature and composition, particularly in heavy oil or bitumen production. The formation of stable emulsions, where one liquid phase is finely dispersed within another (e.g., oil-in-water or water-in-oil emulsions), creates a nearly homogenous mixture that optical or electrical sensors may struggle to differentiate, especially if the densities of the emulsified phases become remarkably similar. Furthermore, the presence of surface-active agents can induce severe foaming, particularly in gas-rich streams. Foam, being a low-density, highly viscous mixture of gas and liquid, can drastically alter the apparent mixture density and impede phase separation, leading to significant measurement inaccuracies and even sensor fouling. The constantly evolving nature of these properties over the life of a well demands that MPFMs possess a high degree of adaptability and robust compensation mechanisms (Klug and Mayinger, 1994).

In summary, accurately measuring multiphase flow demands solutions that transcend single-variable sensing. The dynamic interplay of flow regimes, the variability of fluid properties, the rigors of the operating environment, and the practical constraints of field deployment collectively conspires to make MPFM one of the most intellectually and technically demanding pursuits in petroleum engineering. Overcoming these challenges necessitates continuous innovation in sensor fusion, sophisticated signal processing, and robust algorithmic development, paving the way for more resilient and intelligent measurement systems.

This paper delves into the field-proven performance of such a next-generation non-nuclear MPFM, showcasing its precision, reliability, and cost-effectiveness. By integrating advanced Sonar, Near Infra-red, and Venturi technologies, this system not only addresses the practical limitations but also sets a new benchmark for environmentally responsible and economically viable hydrocarbon production in the digital age. The engineering solution explores how these technological advancements are reshaping the industry, offering a glimpse into a future where real-time, high-fidelity data empowers smarter decisions, minimizes environmental footprints, and unlocks the full potential of every producing asset.

Non-Nuclear Multiphase Flow Measurement: Technology and Engineering Design

The multiphase flow meter deployed in this study represents a significant advancement in hydrocarbon measurement, offering full-range, non-nuclear capabilities with enhanced data intelligence and operational simplicity. This technology is engineered for versatile production environments, including heavy oil and wet gas applications, and accurately monitors individual or group wells in real-time. By eliminating the necessity for costly test separators and circumventing the complex nuclear-source management associated with conventional MPFMs, this system significantly reduces both capital and operating expenses while concurrently improving well-test frequency and data accuracy.

The system's core design integrates three distinct, yet complementary, technologies: Sonar, Near Infrared (NIR), and a Venturi meter, alongside an intelligent flow computer. This multi-sensor approach enables precise characterization of multiphase flow across a broad spectrum of gas-to-liquid ratios (GLRs) without requiring external separation or a gamma densitometer.

System Components and Measurement Principles

Sonar Technology

The Sonar component leverages the inherent acoustic properties of turbulent pipe flow as shown in Figure 2. Produced fluids generate self-sustaining turbulent eddies that propagate along the pipe. An array of axially distributed pressure sensors precisely measures the dynamic pressure fluctuations associated with these eddies. Phased array processing algorithms analyze these patterns to calculate the bulk flow velocity. This bulk volumetric flow rate is directly proportional to the convective velocity measured by Sonar. For single-phase flows, a single Reynolds number calibration typically yields a flow rate with an uncertainty better than $\pm 1\%$. In wet gas configurations, Sonar serves as the primary total flow rate measurement. Within a multiphase flow context, it functions as the total flow rate measurement at high flow rates and provides a crucial cross-check for the Venturi nozzle at normal operating conditions. Additionally, an array of external strain sensors contributes data for gas, liquid, and mixed phases, which is processed by a proprietary algorithm requiring no recalibration.

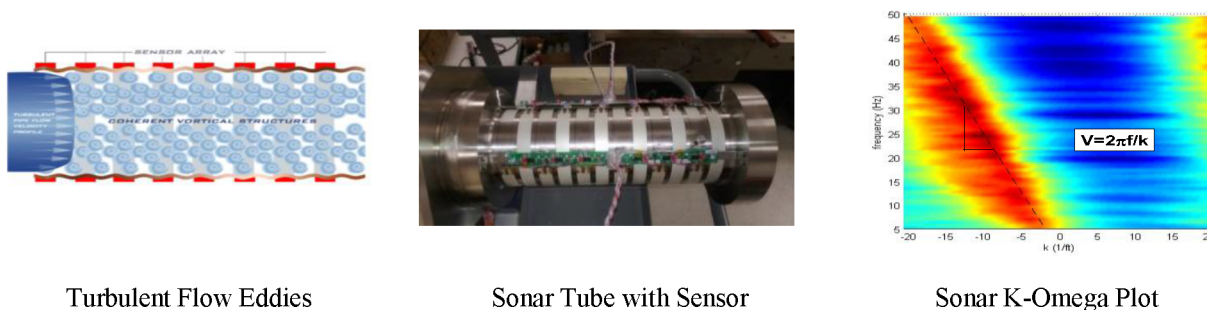


Figure 2—Sonar Technology

Near Infrared (NIR) Technology

The NIR sensor (Figure 3) provides a full-range 0-100% water cut measurement, accommodating the wide variety of conditions encountered in the oilfield. This sensor operates independently of brine salinity, liquid density variations, and is notably insensitive to entrained gas. The Meter is founded on the principle of near-infrared absorption. Oil and water possess distinct absorption spectra in the NIR region. By meticulously measuring light absorption at key wavelengths, the meter accurately quantifies the relative concentrations of oil and water, irrespective of free gas content. Crucially, this absorption measurement targets the molecular

structure of water and oil, rather than dissolved components like salt, ensuring robust accuracy regardless of water chemistry.



Figure 3—Near Infrared (NIR) Technology

Venturi Meter

The Venturi meter, a differential pressure device (Figure 4), provides a measurement directly related to the flow momentum of the multiphase mixture. This enables the determination of either fluid density or volumetric flow rate. In scenarios where an independent measurement of volumetric flow rate is available, the Venturi can output the mixture density by applying the momentum balance, assuming homogeneous flow within the nozzle. A linear dependency between the fluid mixture density and the gas volume fraction (GVF) can be established. In the wet gas configuration, the Venturi nozzle is utilized to measure GVF, whereas in the multiphase configuration, it serves as a primary input for total flow rate determination.

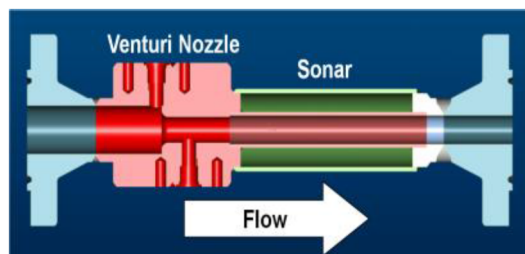


Figure 4—Venturi-Sonar Schematic

Integrated Flow Computer

The flow computer is the central processing unit, retrieving sensor data every second to calculate real-time flow rates and accumulated values over user-configured intervals. Flow rate values, along with lower-level calculated values and diagnostic parameters, are accessible on-screen and via Modbus. Logged data can be retrieved by physically swapping the storage device card or accessing it remotely via the Ethernet port, requiring no special monitoring software or licenses.

The seamless combination of Sonar for velocity and gas fraction, NIR for oil/water fraction, and the Venturi meter for total flow rate empowers this non-nuclear MPFM to deliver comprehensive and accurate individual phase flow rates, effectively eliminating the need for radioactive sources.

Iterative Mass Balance Algorithm for Multiphase Flow Metering

In multiphase flow metering, accurately determining the individual flow rates of oil, water, and gas from a commingled stream necessitates sophisticated algorithmic approaches that reconcile disparate sensor inputs. Figure 5 illustrates a fundamental mass balance iteration workflow employed by an MPFM integrating differential pressure (DP) and Sonar measurements, anchored by a PVT (Pressure, Volume, Temperature)

fluid model. At its core, this workflow iteratively refines the estimated gas/liquid (G/L) fraction to achieve a converged, mass-balanced solution for the total flow. The process is initiated by feeding measured system pressure (P) and temperature (T) into a PVT model, which precisely calculates the real-time densities and volumetric properties of oil, water, and gas at flowing conditions. Concurrently, raw differential pressure (DP) data from, for instance, a Venturi meter, is processed by a DP model, which, leveraging the fluid properties and an initial G/L fraction estimate, calculates a DP mass rate. In parallel, Sonar measurements, representing total mixture velocity (V), are processed by a Sonar model to yield a Sonar mass rate, similarly utilizing the fluid properties and the current G/L Fraction estimate. The calculated DP mass rate and Sonar mass rate are then rigorously compared in a mass balance check. If a significant discrepancy exists, indicating a deviation from mass conservation based on the current G/L fraction, an "Update G/L fraction" step adjusts this critical parameter. This refined G/L fraction is fed back into both the DP and Sonar models, initiating another iteration. This cycle continues until the total mass rates derived from both DP and Sonar measurements converge within an acceptable tolerance. Upon convergence, the system has effectively determined the true G/L fraction, enabling the MPFM to accurately partition the total mass flow into the individual mass (and subsequently volumetric) flow rates of each phase, providing a robust and reliable solution for complex multiphase flow characterization.

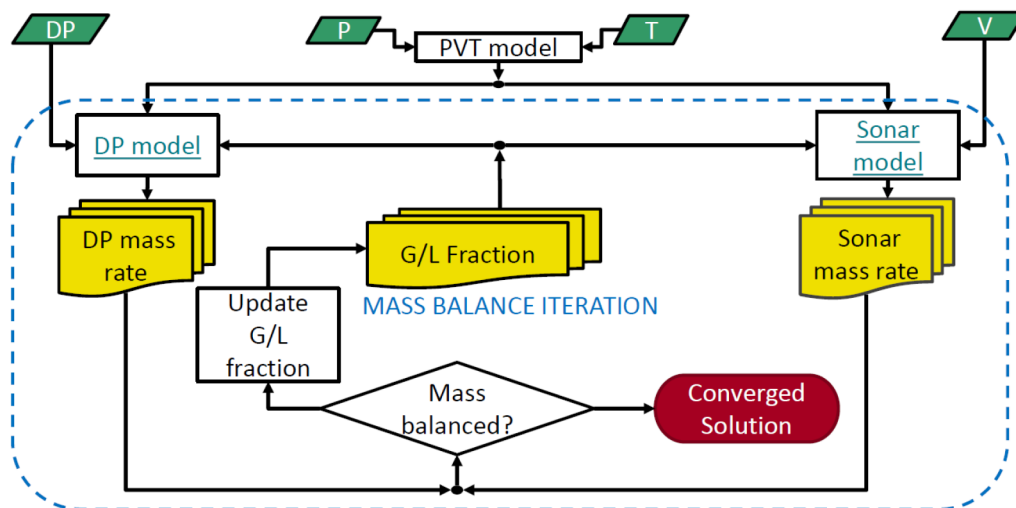


Figure 5—Mass Balance Iterative Calculations

Methodology Workflow for MPFM Verification Against a Test Separator

The accurate verification of a multiphase flow meter against a traditional test separator is a critical step in establishing the MPFM's reliability and proving its commercial and operational viability. This rigorous process, often conducted as a field trial or site acceptance test, aims to confirm the MPFM's measurement accuracy for individual oil, water, and gas flow rates under actual well conditions. The methodology typically follows a structured workflow to ensure comprehensive data collection, comparison, and analysis. The verification process involves running the MPFM in series with a conventional test separator, allowing for simultaneous measurement of the same well stream by both systems. This comparative approach provides the necessary data to assess the MPFM's performance against a test separator reference.

Conventional test separators (Figure 6), a long-standing fixture in the oil and gas industry, primarily rely on gravity for the physical separation of fluid phases. While robust in principle, these traditional separators can encounter significant challenges, particularly when dealing with difficult fluid conditions such as stable emulsions or foamy mixtures. Such conditions can severely hinder effective phase separation, directly leading to inaccuracies in the measured individual flow rates of oil, water, and gas.

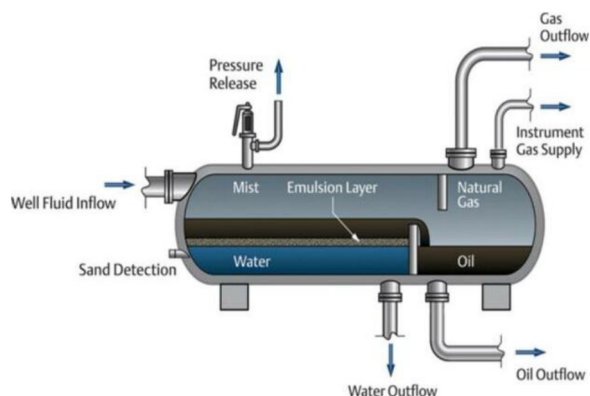


Figure 6—Oil and Gas Separator (After Scapin et al., 2017)

Furthermore, a notable operational distinction lies in their adaptability: traditional separators often necessitate recalibration or adjustments for specific fluid compositions to maintain measurement integrity. In contrast, advanced multiphase flow meters MPFM as shown in Figure 7 demonstrate a superior capability to adapt to varying flow conditions with minimal or no recalibration. This inherent flexibility allows MPFMs to measure flow rates across a significantly broader range of fluid compositions and operational parameters without requiring extensive manual adjustments. Such adaptability proves particularly critical in dynamic production environments, such as tight carbonate plays, where fluid characteristics and well performance can evolve frequently over time. Beyond operational flexibility, MPFMs offer a substantial advantage in terms of physical footprint. Their compact design eliminates the need for large separation vessels, enabling easier and more cost-effective installation in space-constrained locations, unlike traditional separators, which typically demand more extensive and often costly infrastructure. However, the test separator is still considered a reliable method and is used as a reference metering technology to cross-check, validate, and compare the MPFM results in oil and gas field trial test for performance accuracy.



Figure 7—Non-radioactive MPFM

Field Setup

In the field setup for MPFM validation, the MPFM was installed upstream of a test separator to enable direct performance comparison as presented in Figure 8. The MPFM continuously measured multiphase flow rates oil, water, and gas based on its internal algorithms and sensor data. Downstream, the test separator physically separated the production stream into individual phases, allowing for accurate, independent measurement of oil, water, and gas rates using conventional single-phase flow meters. This arrangement ensured that both

MPFM and separator were exposed to the same well stream conditions, enabling effective validation of the MPFM's phase rate predictions against the test separator's benchmark measurements.

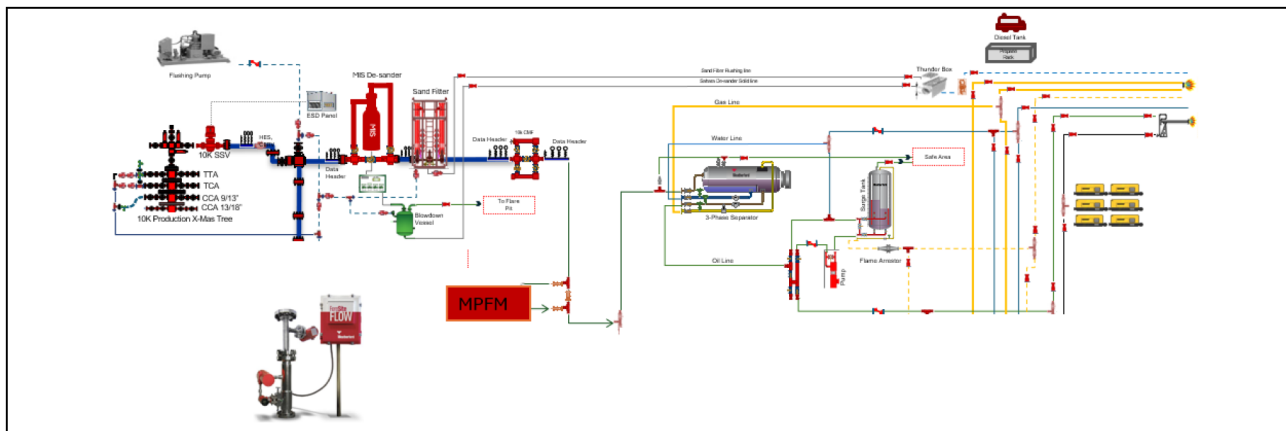


Figure 8—Field Setup for MPFM versus Test Separator

Test Results and Discussion

This section presents the comprehensive results obtained from the field verification of the non-nuclear multiphase flow meter against a conventional test separator. The primary objective of this testing was to validate the MPFM's measurement accuracy and reliability under actual well operating conditions, thereby confirming its suitability for continuous production monitoring and allocation. The methodology involved simultaneous data acquisition from both the MPFM and the operated test separator in terms of oil rate, water rate, gas rate, and water cut %. These results present two MPFM field trials that were executed across distinct unconventional wells to evaluate performance under varying flow regimes Well-1 and Well-2.

Well-1, an unconventional well, presents significant challenges due to its volatile oil production and highly complex, unstable multiphase flow. The well stream yields a fluid stream composed of light sour crude, associated gas, formation water, and notable concentrations of hydrogen sulfide (H_2S). Under surface multiphase flow meter (MPFM) conditions, Well-1 exhibits predominantly high gas volume fraction (GVF) oil flow scenarios, with GVF values reaching as high as 98%. This high GVF indicates a system dominated by gas, even in what is primarily an oil-producing well, contributing to the instability and complexity of the flow behavior. The presence of H_2S further complicates operations due to its corrosive nature and safety implications, necessitating specialized handling and processing.

As shown in Figure 9 over the monitoring period, inline pressure initially holds at a high level before stepping down and then gradually declines with minor fluctuations, while temperature remains consistently high with only a slight decrease toward the end. The differential pressure mirrors this behavior in a clear staircase pattern dropping sharply, plateauing, and then dropping again to lower levels. Meanwhile, the gas volume fraction exhibits several peaks and troughs but trend modestly downward overall. Together, these trends indicate a transition to lower pressure and differential pressure conditions, stable high-temperature operation, and an increasingly gas-rich, yet variable, flow regime.

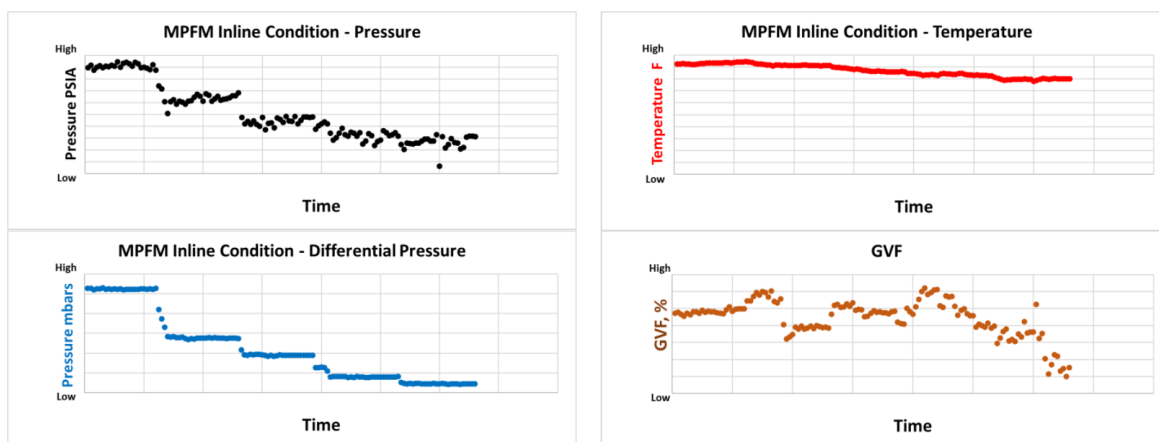


Figure 9—Well-1 MPFM Inline Condition Pressure, Temperature, Differential Pressure, GVF

Well-2 is an unconventional gas-condensate well that produces a complex, sour mixed-phase stream dominated by very high gas-oil ratios, light condensate and co-produced water, with moderate H_2S (and CO_2) content. At the surface MPFM point it behaves as a wet gas application with high gas volume fractions necessitating precise phase-split control, advanced slug-mitigation and rigorous flow-assurance strategies. The accurate multiphase metering and dynamic pressure-control are critical to optimize recovery, ensure operational integrity and maintain safety standards.

Figure 10 shows Well-2's MPFM inline conditions over testing time. The multiphase flow meter performance under wet gas conditions demonstrates characteristic operational trends. System pressure shows a gradual increasing trend with intermittent surges, while temperature remains remarkably stable throughout the monitoring period. The differential pressure profile exhibits an initial buildup followed by a sharp decline to minimal values, indicating either a transition to single-phase flow or effective liquid slug clearance. The consistently high gas volume fraction confirms sustained wet gas flow.

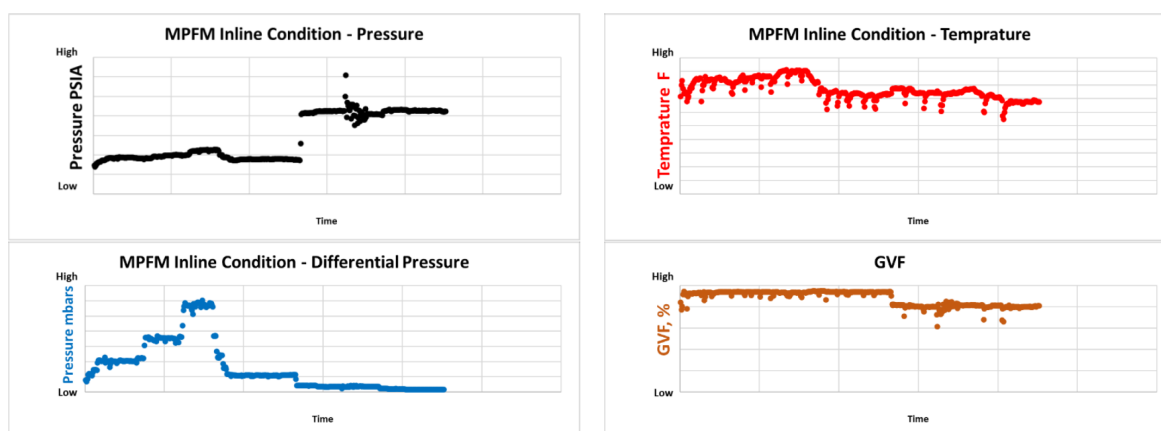


Figure 10—Well-2 MPFM Inline Condition Pressure, Temperature, Differential Pressure, GVF

Field Trial Test Results:

MPFM performance was evaluated with respect to the test separator over the period of several weeks. The test results were observed as below.

The MPFM Well-1 test (Figure 11) demonstrates strong agreement with separator measurements, particularly for gas flow rate (MMSCFD) and water-liquid ratio (WLR%), both showing a consistent and low 10% error, confirming reliable performance in these critical parameters. While liquid flow rate (BBL/D) measurements exhibit slightly higher variability (11.9–12.3% error) indicates stable measurement precision.

These results validate the MPFM's robustness in multiphase flow conditions, with gas and WLR accuracy matching separator performance as a key achievement for field deployment.

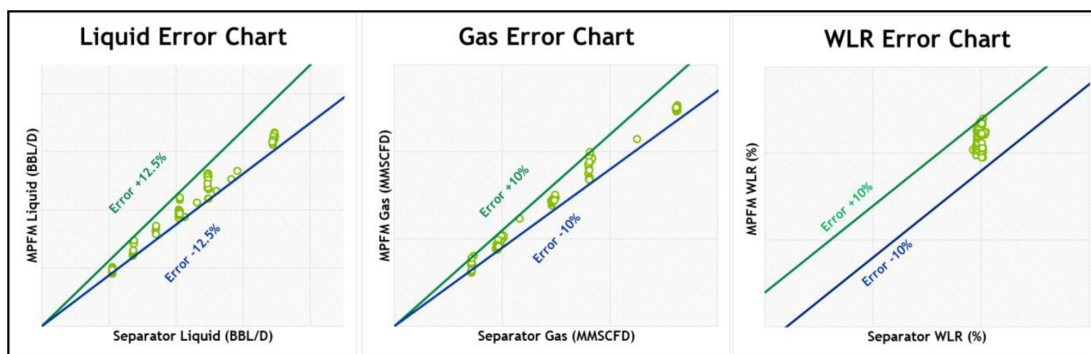


Figure 11—Well-1 Results

The MPFM Well-2 test (Figure 12) demonstrates excellent measurement consistency, with gas (MMSCFD) and water-liquid ratio (WLR%) errors matching separator performance at a low +10% across all readings. While liquid measurements (BBL/D) show slightly higher errors (+12.5% vs separator's), the identical repeated values for both MPFM and separator measurements indicate highly stable and repeatable performance. The results confirm the MPFM's reliability for gas and WLR measurements, with liquid measurements showing consistent. The data particularly validates the system's precision in gas and water-cut determination, which are critical for reservoir management and production optimization.

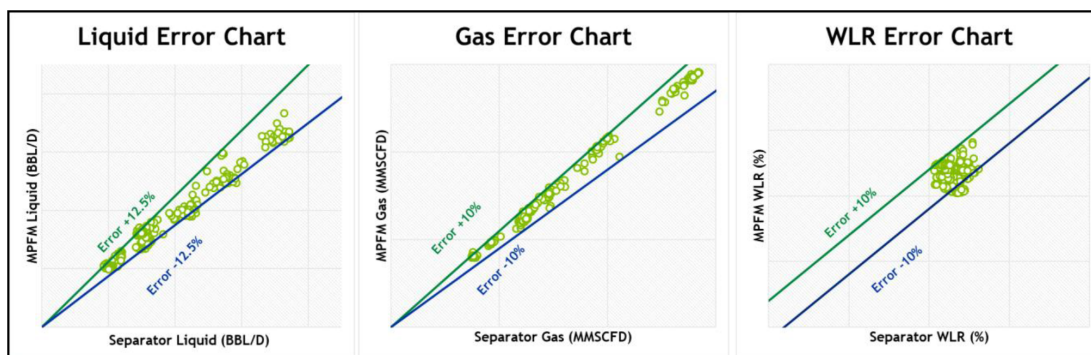


Figure 12—Well-2 Results

The test results illustrated in the above plots demonstrate that the MPFM measurements for liquid flow rate, gas flow rate, and water liquid ratio are in close agreement with the reference Test Separator data and fall within the defined acceptance criteria. Based on this performance, the MPFM field trial is considered successful.

Technology Added Value: Enhancing Sustainability and Profitability

Advanced measurement technologies, such as MPFMs, deliver substantial added value across operational, environmental, and economic domains within industrial applications, notably in the oil and gas sector. These systems offer significant advantages over conventional methods, enhancing efficiency, reducing environmental impact, and improving financial performance.

Advanced measurement systems are pivotal for achieving operational excellence by providing real-time, continuous performance data. This immediate visibility into key process parameters, like individual fluid flow rates and compositions, empowers engineers to make faster, data-driven decisions for optimization

and intervention. Continuous monitoring enables the early detection of anomalies, facilitating proactive responses that minimize downtime and prevent operational issues from escalating. Furthermore, these technologies enhance efficiency by reducing the need for extensive manual intervention and periodic testing, freeing personnel for higher-value tasks and improving safety. Their compact and integrated design simplifies installation, lowers maintenance, and reduces overall operational complexity, ensuring consistent data quality and fewer disruptions.

From an environmental standpoint, these technologies play a crucial role in reducing industrial footprints. By providing precise, continuous data, they enable optimized resource utilization, maximizing recovery while minimizing waste. Compared to conventional methods that might involve venting or flaring, the reduced reliance on large-scale physical separation contributes to lower direct operational emissions. The inherent design often features fewer mechanical connections and potential leak paths, significantly reducing fugitive emissions of hydrocarbons and other fluids. Moreover, the smaller overall physical footprint of these advanced systems, across their entire lifecycle from manufacturing to operation, results in a lower environmental burden. MPFMs deliver compelling environmental and economic benefits, with studies showing they can reduce CO₂ emissions by 67% compared to conventional separators (Plagens et al., 2018). This makes them a cleaner solution for production monitoring while maintaining measurement accuracy.

Economically, innovative measurement solutions offer compelling advantages throughout a project's financial lifecycle. While not directly revenue-generating, their impact on cost reduction and efficiency is significant. The Capital Expenditure (CAPEX) for these advanced systems is typically lower than for traditional, larger-scale infrastructure. Crucially, Operating Expenditures (OPEX) are also reduced due to lower energy consumption, decreased maintenance frequency, and optimized labor requirements. These combined savings in both capital and operating costs notably improve overall project economics, contributing to a more favorable Net Present Value (NPV) by minimizing expenditures and optimizing resource allocation over the project's lifespan. Financially, MPFMs offer superior value, with CAPEX projected to be 40–50% lower and OPEX 25–30% less than traditional systems. Field data from the Eagle Ford Basin confirms these savings, demonstrating over 50% reductions in both CAPEX and OPEX while achieving higher NPV making MPFMs both cost-effective and sustainable for modern hydrocarbon operations (Almurayshid et al., 2024).

Frontiers in Multiphase Flow Metering: R&D Advancements

From the research and development (R&D) perspective, multiphase flow metering (MPFM) is a vibrant and critical field within the oil and gas industry, constantly pushing the boundaries of measurement science. The core objective remains the accurate and continuous quantification of individual oil, water, and gas flow rates in a commingled stream, but the methods and underlying intelligence are undergoing significant transformation. A major thrust in R&D is the innovative sensor design and sophisticated data interpretation. Researchers are heavily invested in providing more granular and accurate cross-sectional information about phase distribution. Concurrently, advancements in microwave and radio frequency (RF) sensors are exploring new frequency ranges and modulation techniques.

Beyond individual sensor enhancements, a transformative area of R&D is the profound integration of Artificial Intelligence (AI) and Machine Learning (ML). AI models learn complex, non-linear correlations between raw, noisy sensor outputs and actual multiphase flow rates, compensating for real-world complexities like slippage between phases and non-ideal flow profiles that are difficult to capture with deterministic models (Bao et al., 2024). A closely related and rapidly evolving R&D domain is virtual flow metering (VFM) as presented in Figure 13. While physical MPFMs provide direct measurements, VFMs offer a cost-effective alternative by estimating well production rates using readily available surface and downhole data combined with sophisticated models. The latest research in VFM is heavily reliant on hybrid

models that blend mechanistic (physics-based) multiphase flow simulators with data-driven ML algorithms (Li et al. 2024).

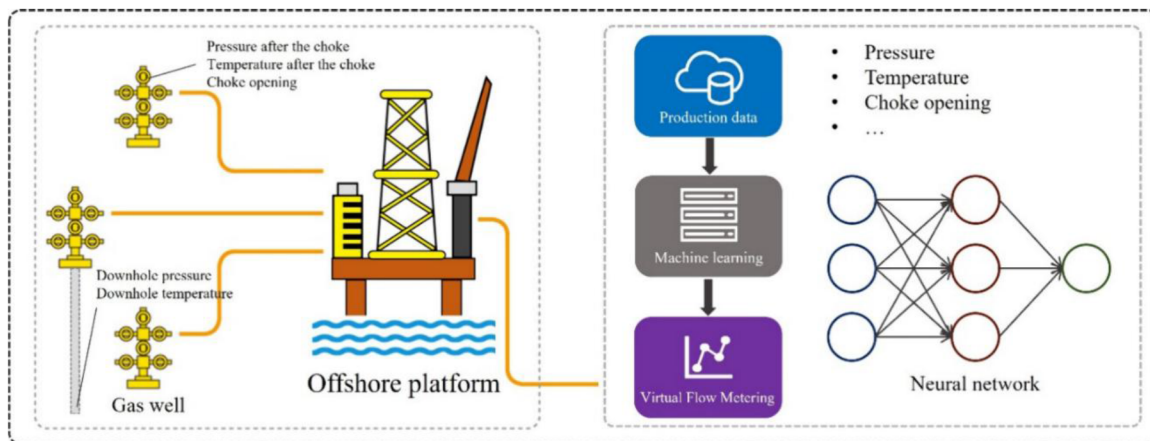


Figure 13—Predictive Model for Virtual Metering (Li et al., 2024)

In essence, R&D in multiphase flow metering is a multi-faceted endeavor driven by the need for more intelligent, autonomous, and environmentally conscious solutions. The convergence of advanced sensor physics, sophisticated computational intelligence, and a holistic approach to flow assurance is defining the next generation of MPFM technologies.

Conclusions

- A comprehensive technical assessment of advanced multiphase flow metering technology confirms its indispensable role in optimizing contemporary hydrocarbon asset management. Addressing the intrinsic complexities of heterogeneous fluid streams, MPFMs, particularly non-nuclear variants leveraging diverse physical principles, demonstrably surmount the limitations of conventional gravity-based phase separation.
- Two fields test results of advanced multiphase flow metering technology confirm its vital role in optimizing hydrocarbon asset management. Field testing demonstrated highly accurate performance, with gas (MMSCFD) and water-liquid ratio (WLR%) measurements matching separator precision at a consistent +10% error margin. While liquid flow (BBL/D) measurements showed slightly elevated but stable variance (+12.5% vs. separator's), the MPFM's repeatability and reliability in gas and WLR quantification is critical for reservoir management.
- This validation confirms the system's metrological precision and operational reliability under actual well conditions, thereby substantiating its efficacy for real-world production surveillance and allocation.
- This technological evolution directly translates into quantifiable economic efficiencies via optimized CAPEX and reduced OPEX, concurrently generating a critical positive environmental impact by mitigating greenhouse gas emissions and minimizing physical infrastructure footprint. Consequently, the strategic adoption of MPFMs represents a foundational pillar for achieving operational excellence, facilitating granular, real-time data-driven reservoir and production management, thereby fostering a more sustainable and thermodynamically efficient petroleum industry.

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